

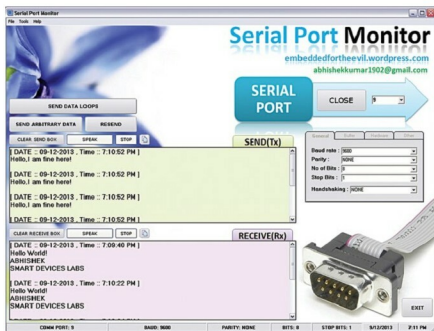


Fig. 2: The main program output window



Fig. 3: Save serial conversation feature

1. **Data carrier detect (DCD).** Determines if the modem is connected to a working phone line
2. **Receive data (RXD).**

Computer receives information sent from the modem

3. **Transmit data (TXD).** Computer sends information to the modem

4. **Data terminal ready (DTR).** Computer informs the modem that it is ready to communicate

5. **Signal ground (GND).** Pin is grounded

6. **Data set ready (DSR).** Modem informs the computer that it is ready to communicate.

7. **Request to send (RTS).** Computer asks the modem if it can send information.

8. **Clear to send (CTS).** Modem informs the computer that it can send information.

9. **Ring indicator (RI).** Once a call has been indicated, computer acknowledges signal (sent from modem) to confirm that a ring is detected.

Voltage sent over the pins can be in

one of two states; 'on' or 'off.' Voltage signal between -3V and -25V is considered on (binary value '1') whilst signal between +3V and +25V is considered off (binary value '0'). The intermediate +3V to -3V levels are not used and thus the noises which usually occur in this range have lesser influence over communication.

An important aspect of serial communications is the concept of flow control. This is the ability of one device to instruct another to stop sending data for a while. The commands such as request to send (RTS), clear to send (CTS), data terminal ready (DTR) and data set ready (DSR) are used to enable flow control. With flow control, the modem stops the flow of data from the computer before it overruns the modem's buffer. The computer constantly sends a signal on the RTS pin and checks for a signal on the CTS pin. If there is no CTS response, the computer stops sending data, waiting for the CTS before it resumes. This allows the modem to keep the flow of data running smoothly.

The RS-232 standard. RS-232 (Recommended Standard-232) is a standard interface approved by the Electronic Industries Association (EIA) for connecting serial devices. This is

the interface that a computer uses to communicate and exchange data with a modem and other serial devices.

Software program

The serial-port monitor software is designed, coded and compiled using the Visual Basic 6.0, which is a rapid application development (RAD) tool designed by Microsoft for developing fast and effective applications for Windows operating system. Visual Basic 6.0 integrated development environment (IDE) offers great tools to easily design the graphical user interface (GUI) for your applications. In general, it is an event-based programming environment where an activity triggers an event and that event is processed by an event handler which executes the programmed code for that particular event. The software uses the 'MSCOMM32.ocx' object to communicate with the serial port at the computer, 'Voice.dll' for text-to-speech function, 'winmm.dll' to play .wav audio files, 'comdlg32.ocx' for common dialog box display, 'tabctl32.ocx' for displaying the multitab control on the GUI, 'richtx32.ocx' for providing rich text boxes on the GUI for received and sent data, 'INIRead' and 'INIWrite' modules to read and write software configuration files (.ini files) for storing data and uses a few more general-purpose objects like command buttons and combo boxes.

Running the application. The main window that appears when program is run is shown in Fig. 2. This is the main work area of the software. Starting from the top, the software has been provided with the File, Tools and Help menu bars where features like saving of data to text file, etc can be accessed. As shown in Fig. 3, in the File menu, the user can click on Save option to save the serial conversation that has taken place in the software. When you click the Save button (or press Ctrl+S), the software will ask you to give a filename to store the conversation (e.g., transmit and receive data) and then will store it into a rich text file format (RTF) with the date and time.

To exit the application, either you